

Noise-Cancelling Air Purifier

Innovation dossier

1. What is it?

A concept for the real 'Noise / motor sound (weak real signal)' friction, built with the CROSS_CATEGORY_TRANSFER design move - transfer a verified capability from another product/category.

2. Physical architecture

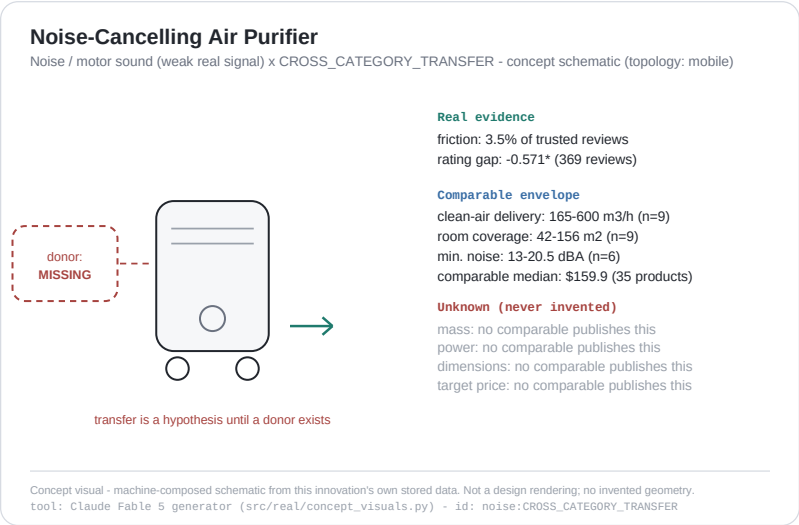


Figure: Concept visual

Machine-composed schematic from the innovation's stored fields only (operator rule, friction stats, comparable envelope). Not a design rendering; a Fable render or drawing would also be CONCEPT_VISUAL, never a digital model or physical prototype.

artifact visual-noise:CROSS_CATEGORY_TRANSFER - /concept-visuals/noise_CROSS_CATEGORY_TRANSFER.svg

Product archetype

- room scale:** Unknown - no real data in this pipeline for this field.
- mobility:** Unknown - no real data in this pipeline for this field.
- size class:** Unknown - no real data in this pipeline for this field.
- power class:** Unknown - no real data in this pipeline for this field.
- noise ambition:** Unknown - no real data in this pipeline for this field.
- maintenance burden:** Unknown - no real data in this pipeline for this field.
- filter service dependency:** Unknown - no real data in this pipeline for this field.
- connected:** Unknown - no real data in this pipeline for this field.
- form factor:** mobile

Design rule - each populated field follows from the operator's own fixed definition, never from evidence

System architecture

Unknown - no system-architecture dataset exists; the archetype above is a design rule from the operator's own definition.

Method choice

CROSS_CATEGORY_TRANSFER - Transfer a verified capability from another product/category.

Method choice - authored design vocabulary

3. Why it exists

Reality Observed

3.5% of trusted reviews (369 reviews across 77 real products) carry this friction, with a -0.571* average rating gap. Signal: Motor/fan noise is a recurring complaint, now backed by peer-reviewed evidence it drives real under-use.

Households whose real reviews carry this friction: 369 reviews across 77 real products (69.6% verified purchase, 2004-10-14-2023-08-08).

Unknown - no persona/behavioural dataset exists in this pipeline; only the review evidence above is real.

Transformation Method choice

METHOD CHOICE, not evidence: the authored operator table assigns CROSS_CATEGORY_TRANSFER - "Transfer a verified capability from another product/category." - to this friction. Analyst design judgment, applied deterministically.

Product consequence Research tension

Optimize for hours-actually-run under real noise tolerance, not peak lab CADR. Assumption challenged: "Clean-air delivery rate (CADR) is the primary performance basis." -> What if performance were reported as a loss-rate curve or realized exposure reduction, not a lab CADR number?

4. Evidence and lineage

Parent paths

tension:T1 - Filtration vs. noise-driven adherence Tension

test: Falsified when one design point achieves RP-01's measured primary-room particulate reduction at a sound level that removes noise as a named reason for running the device less (RP-05) - both halves are stored, quantified findings.

derived possibilities: noise:AMBIENT, noise:TEMPORAL_SHIFT, noise:CROSS_CATEGORY_TRANSFER, value_effectiveness:CONCENTRATE, value_effectiveness:CROSS_CATEGORY_TRANSFER, value_effectiveness:PERSONALISE

tension:T2 - Constant vs. automatic operation Tension

test: Falsified when an auto-mode arm reaches the lower bound of RP-07's constant-mode reduction interval while keeping auto's runtime/noise advantage - the two stored confidence intervals would then overlap and the trade-off dissolves.

derived possibilities: noise:AMBIENT, noise:TEMPORAL_SHIFT, noise:CROSS_CATEGORY_TRANSFER, value_effectiveness:CONCENTRATE, value_effectiveness:CROSS_CATEGORY_TRANSFER, value_effectiveness:PERSONALISE

tension:T3 - CADR theory vs. real-home performance Tension

test: Falsified when a CADR-derived sizing prediction matches a source-strength-adjusted measured loss rate within the fan-speed band RP-06 reports, in a home with the central air-handling configuration RP-01 identifies as decisive.

derived possibilities: noise:AMBIENT, noise:TEMPORAL_SHIFT, noise:CROSS_CATEGORY_TRANSFER, value_effectiveness:CONCENTRATE, value_effectiveness:CROSS_CATEGORY_TRANSFER, value_effectiveness:PERSONALISE

tension:T6 - Pollutant reduction vs. clinical outcome certainty Tension

test: Resolved when an independent replication of RP-08 reproduces its clinical effect in a cohort whose particulate reduction matches RP-03's measured range - RP-08's own record names independent replication as the missing step. A failed replication resolves it the other way.

derived possibilities: noise:AMBIENT, noise:TEMPORAL_SHIFT, noise:CROSS_CATEGORY_TRANSFER

assumption:A5 - Clean-air delivery rate (CADR) is the primary performance basis.

Assumption to test

test: Challenged when a certification body in the trend corpus adopts a loss-rate or realized-exposure metric alongside CADR - concretely, when TC-R03 (AHAM Verifide, the one record whose theme tags contain 'cadr': ['certification', 'cadr', 'iaq_standards']) stops centring on it, or TC-R02 (ENERGY STAR, themes ['regulation', 'energy_efficiency', 'iaq_standards'], CADR present only in its prose scope note) adds a non-CADR performance theme tag. Fragile baseline: both documents carry published_date null, so change detection has no dated anchor.

derived possibilities: noise:AMBIENT, noise:TEMPORAL_SHIFT, noise:CROSS_CATEGORY_TRANSFER, value_effectiveness:CONCENTRATE, value_effectiveness:CROSS_CATEGORY_TRANSFER, value_effectiveness:PERSONALISE

assumption:A6 - Continuous, constant operation is the default expectation.

Assumption to test

test: Challenged when connected+adaptive share exceeds 25% of the corpus (currently 23/237 = 9.7%), or when an independent trial reproduces RP-07's constant-mode filtration under auto operation.

derived possibilities: noise:AMBIENT, noise:TEMPORAL_SHIFT, noise:CROSS_CATEGORY_TRANSFER, value_effectiveness:CONCENTRATE, value_effectiveness:CROSS_CATEGORY_TRANSFER, value_effectiveness:PERSONALISE

evidence ids: taxonomy:noise, RP-05, RP-07, TC-R11

category assumptions: A5, A6

Field evidence behind these paths (titles)

- Self-reported health impacts of do-it-yourself air cleaner use in a smoke-impacted community
- Real-World Effectiveness of Portable Air Cleaners in Reducing Home Particulate Matter Concentrations
- Reduction of personal PM2.5 exposure via indoor air filtration systems in Detroit: an intervention study
- The impact of portable air cleaners on indoor particulate matter concentrations and perceptions of indoor air quality: a randomized crossover trial in three multifamily buildings
- Analysis of sources and sinks of indoor particulate matter reveals insights into the real-world efficacy of portable air cleaners in a randomized intervention trial
- Indoor environmental health and asthma relief: findings from a longitudinal HEPA filter trial
- Effectiveness of portable HEPA air cleaners on reducing indoor PM2.5 and NH3 in an agricultural cohort of children with asthma: A randomized intervention trial

Design DNA

F consumer friction (taxonomy:noise) Present

Real Amazon review-text friction theme (see Consumer Pain methodology).

O design operator Method choice

Authored operator vocabulary applied deterministically (METHOD_CHOICE, not evidence) - see OPERATORS/THEME_OPERATORS in src/real/magic_box_real.py.

S signal (noise) Present

Motor/fan noise is a recurring complaint, now backed by peer-reviewed evidence it drives real under-use.

T scientific tension Present

Filtration vs. noise-driven adherence; Constant vs. automatic operation; CADR theory vs. real-home performance; Pollutant reduction vs. clinical outcome certainty

R rival gap Present

11 named real competitors measurably weaker on this theme. [Dyson, Plixio, Oreck, Envion, Wynd, Molekule, AMEIFU, AZEUS, AIRTHEREAL, Egofine, Elechomes]

C versuni capability Missing / unverified

No real Versuni internal capability/org-readiness dataset exists in this pipeline.

A category assumption Present

Clean-air delivery rate (CADR) is the primary performance basis.; Continuous, constant operation is the default expectation.

E economic condition Present

Real price-weighted exposure computed for this theme (\$84,015.62).

5. Comparable products

comparable set: 10 verified official Versuni/Philips air-purifier families (room-scale)

clean-air delivery rate (CADR): 165-600 m3/h across 9 real comparables
source: individually verified official Versuni/Philips product pages (data/visual/product_images.json)

room coverage: 42-156 m2 across 9 real comparables
source: individually verified official Versuni/Philips product pages (data/visual/product_images.json)

minimum noise: 13-20.5 dBA across 6 real comparables
source: individually verified official Versuni/Philips product pages (data/visual/product_images.json)

6. Engineering envelope

metric	value from verified comparables	epistemic type
clean-air delivery rate (CADR)	165-600 m3/h (n=9 comparables)	Observed comparable
room coverage	42-156 m2 (n=9 comparables)	Observed comparable
minimum noise	13-20.5 dBA (n=6 comparables)	Observed comparable
target mass	No verified comparable publishes this value - left unknown, never estimated.	Unknown
target power	No verified comparable publishes this value - left unknown, never estimated.	Unknown
target dimensions	No verified comparable publishes this value - left unknown, never estimated.	Unknown

7. Economics

price-weighted exposure: \$84015.62

SUM of real observed listed prices across affected reviews with a known price. This is a RELATIVE exposure indicator, not a revenue or market-size estimate - it has no units-sold, conversion-rate, or time-period basis. It answers 'which friction touches more expensive products' not 'how much money is at stake per year.'

comparable market median: \$159.9 (n=35 products) - not this concept's price

MEDIAN real listed price across the 35 distinct real products affected by this friction that have a known price. This is what products in this segment actually cost today - not a proposed price for a new concept, which this evidence cannot establish.

target price: Not modelled - No concept target-price model exists - only the comparable market median (\$159.9, 35 real products) as reference. A target price would be invented.

8. Prototype

State: Concept visual

Concept visual means a schematic exists, nothing more: no digital model, no physical prototype, no test data.

Machine-composed schematic from the innovation's stored fields only (operator rule, friction stats, comparable envelope). Not a design rendering; a Fable render or drawing would also be CONCEPT_VISUAL, never a digital model or physical prototype.

9. Experiment and next learning

Next experiment

A real acoustic-engineering standard or published low-RPM motor precedent, if found, would raise this to high; a real internal R&D estimate showing it requires novel hardware would lower it to low.

Kill criterion

No authored kill criterion yet for this innovation - only the formal case (see formal_case.abandon_signal) has a human-authored one; this innovation earns its own by developing further evidence or by a human decision to invest.

10. What could kill it?

Where this idea stands: Developing

Non-dominated and grounded in a real research tension.

kill criterion: No authored kill criterion yet for this innovation - only the formal case (see formal_case.abandon_signal) has a human-authored one; this innovation earns its own by developing further evidence or by a human decision to invest.

Uncertainties

- No real acoustic-engineering trend document exists in this corpus at all - the medium rating is a judgment call made in the absence of any external precedent, not a citation.

Critic dimensions

Evidence	Survive	Real Consumer Pain evidence: prevalence 3.5%, average rating gap -0.571 - materially above the gate floor.
Economic	Survive	Real price-weighted exposure (\$84,016) is at or above the corpus median (\$16,113).
Competitive	Survive	Real white space: 11 named real rivals are measurably weaker on this theme.
Human	Needs evidence	No real usability/human-factors evidence exists in this pipeline for this concept.
Physical	Needs evidence	No real lab/engineering feasibility data exists in this pipeline for this concept.
Versuni fit	Needs evidence	No real organizational-capability assessment exists in this pipeline for this concept.
Timing	Needs evidence	No real market-timing/window evidence exists in this pipeline for this concept.
Robustness	Needs evidence	No real sensitivity/robustness study exists in this pipeline for this concept.

critic overall: Needs evidence

Criteria results

Needs evidence: 32 - Pass: 13 - N/A: 4 - Challenge: 3

B5	Challenge	Filtration vs. noise-driven adherence; Constant vs. automatic operation; CADR theory vs. real-home performance; Pollutant reduction vs. clinical outcome certainty
P2	Challenge	Filtration vs. noise-driven adherence; Constant vs. automatic operation; CADR theory vs. real-home performance; Pollutant reduction vs. clinical outcome certainty
F3	Challenge	medium

Lineage and provenance

generated by: src/real/innovations_real.py

magic run input sha256: 0b3f9862e0e270a295eacd709182802e981090e817f4f1072000e37aa83f150c

funnel snapshot: d5d03b0d3856721f04d7d1c4ee7f2fe2a7d1df64ac4d2d4130bede07d2b13f61

figure source: /concept-visuals/noise_CROSS_CATEGORY_TRANSFER.svg (generated 2026-08-31T18:00:05.650001+00:00 by Claude Fable 5 generator - src/real/concept_visuals.py)

data source: data/processed/innovations_real.json + data/processed/funnel_real.json (deterministic; no invented fields)

dossier generator: src/real/generate_innovation_dossiers.py